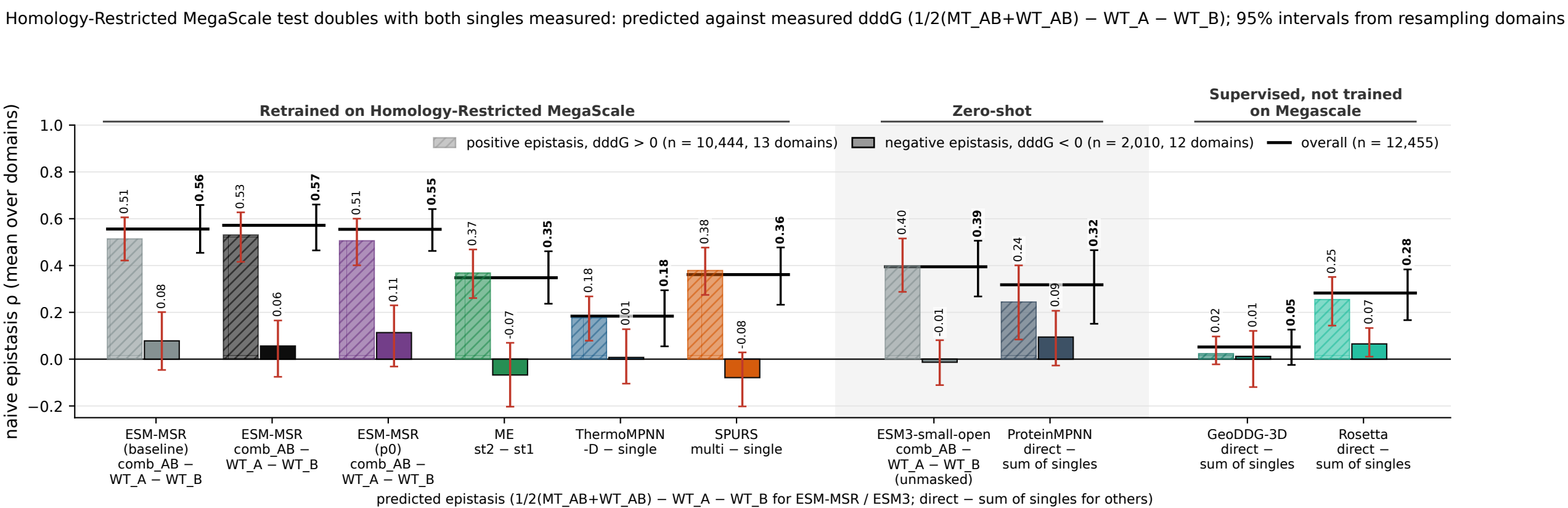
Homology-Restricted MegaScale test doubles with both singles measured: predicted against measured dddG (1/2(MT_AB+WT_AB) − WT_A − WT_B); 95% intervals from resampling domains
Retrained on Homology-Restricted MegaScale
Zero-shot
Supervised, not trained on Megascale
positive epistasis, dddG > 0 (n = 10,444, 13 domains)
negative epistasis, dddG < 0 (n = 2,010, 12 domains)
overall (n = 12,455)
naive epistasis ρ (mean over domains)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
0.51
0.08
0.56
0.53
0.06
0.57
0.51
0.11
0.55
0.37
-0.07
0.35
0.18
0.01
0.18
0.38
-0.08
0.36
0.40
-0.01
0.39
0.24
0.09
0.32
0.02
0.01
0.05
0.25
0.07
0.28
ESM-MSR (baseline) comb_AB − WT_A − WT_B
ESM-MSR comb_AB − WT_A − WT_B
ESM-MSR (p0) comb_AB − WT_A − WT_B
ME st2 − st1
ThermoMPNN -D − single
SPURS multi − single
ESM3-small-open comb_AB − WT_A − WT_B (unmasked)
ProteinMPNN direct − sum of singles
GeoDDG-3D direct − sum of singles
Rosetta direct − sum of singles
predicted epistasis (1/2(MT_AB+WT_AB) − WT_A − WT_B for ESM-MSR / ESM3; direct − sum of singles for others)